
Predicting Pivotal Tokens Before They Are Generated

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Abstract

[TODO: Write last.] Skeleton of the argument: Pivotal Token Search [Abdin et al., 2024, Sharma, 2025] identifies the individual tokens at which a language model’s probability of reaching a correct answer shifts sharply, but it does so *offline* and at a cost of tens of sampled rollouts per candidate position. We ask whether pivotality is recoverable from the model’s own residual stream *one step before* the token is emitted, which would convert an expensive analysis procedure into a single dot product usable during decoding. We further ask whether the *sign* of the impending shift—whether the next token will help or hurt—is separately decodable, a distinction the original pivotality criterion discards by construction.

1 Introduction

[TODO: Write after Phase 2 results land.] Claim ladder to defend, in dependency order (see also Section 5):

1. **Decodability.** Pivotality at step t is linearly decodable from the residual stream at $t - 1$, above chance and above a token-identity control.
2. **Sign.** The *direction* of the impending probability shift is separately decodable at $t - 1$.
3. **Beyond uncertainty.** Both probes exceed next-token entropy, top-2 margin, and Jensen–Shannon baselines.
4. **Amortization.** The probe approximates the PTS oracle at a measured compute ratio.
5. **Selective causality.** Probe-gated intervention along the signed direction, evaluated at matched intervention budget.

2 Background and Related Work

Pivotal tokens. Abdin et al. [2024] introduce Pivotal Token Search to build preference pairs for Phi-4; Sharma [2025] released an open implementation and datasets. [TODO: state the $|\Delta p| > \tau$ criterion formally here.]

Sentence-scale reasoning importance. Thought Anchors [Bogdan et al., 2025] identifies high-influence *sentences* by counterfactual resampling. We work at token scale, which is where the $t - 1$ prediction problem is well posed.

Uncertainty as the incumbent notion of criticality. Wang et al. [2025] show that high-entropy “forking” tokens carry most of the learning signal in RLVR. This makes next-token uncertainty the baseline any claim about token criticality must beat, and we treat it as such rather than as a footnote. Note that entropy is sign-blind by construction, so it cannot in principle explain a working signed probe.

Probing and steering. Linear probes on intermediate representations [Alain and Bengio, 2018, Marks and Tegmark, 2023]; contrastive activation addition [Rimsky et al., 2024, Huang, 2024]; conditional/gated steering [Lee et al., 2024]. Our gate differs in *what it conditions on*: not a semantic category of input, but a prospective, outcome-grounded property of the very next token.

Value versus leverage. Process reward models and value probes [Li et al., 2024] estimate $P(\text{success} \mid \text{prefix})$ —a level. Pivotality is $|\Delta p|$ at the next token—a sensitivity. We include a value-probe-derived baseline to make the distinction empirical rather than rhetorical.

3 Method

3.1 Labels

[TODO: Formal statement.] Group PTS events by source question; the canonical token sequence is the full rollout. Position $t - 1$ immediately preceding a pivotal token is labelled positive; sampled non-pivotal answer-span positions are labelled negative; all other positions are excluded. Extraction at $t - 1$ rather than t is what makes the probe usable during decoding.

3.2 Label noise in the PTS oracle

$\widehat{\Delta p}$ is a difference of two binomial estimates from S rollouts each, accepted when $|\widehat{\Delta p}| \geq \tau$. Under the null, $\text{Var}(\widehat{\Delta p}) = 2p(1 - p)/S$, so at the reference setting $S=50$, $\tau=0.2$ the threshold sits at only 2σ . [TODO: report measured false-positive rate and the accuracy gap between all-labels and the high-confidence subset.]

3.3 Probes

[TODO: Probe 1 (unsigned) and Probe 2 (signed) definitions, regularization, nested layer selection, cluster-bootstrap CIs over questions.]

3.4 Intervention

[TODO: Signed direction $\mu_+ - \mu_-$; cascade gate $p_{\text{steer}} = P_1 \cdot (1 - P_2)$; matched fire rate and matched injected energy against always-on and random-direction controls.]

4 Experimental Setup

[TODO: Fill in once Phase 4 lands.] Models: Qwen3-0.6B/1.7B/4B [Qwen Team, 2025] and DeepSeek-R1-Distill-Qwen-1.5B [DeepSeek-AI, 2025]. Task: GSM8K [Cobbe et al., 2021]. PTS generation uses a vLLM [Kwon et al., 2023] rollout backend.

5 Results

[TODO: Nothing here yet. No number enters this section until it has been produced by the rebuilt pipeline and passed its phase gate.]

6 Limitations

[TODO: Carry these honestly.] Anticipated: PTS labels are themselves noisy (Section 3.2); the pivotality criterion selects for questions in a mid-difficulty band, so pivot density is confounded with task difficulty relative to model capability; results are on a single task family; linear probes test linear decodability only.

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A Appendix

[TODO: Layer sweeps, baseline battery detail, hyperparameters.]